Rithvik Vadlamani

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EDUCATION

The University of Texas at Austin B.S. in Electrical and Computer Engineering May 2028
Coursework: Circuit Theory, Digital Logic Design, Computer Architecture, Embedded Systems, Software Design & Implementation I / II

TECHNICAL SKILLS

Hardware Design: KiCad, schematics, PCB layout, differential pair routing, impedance targets, ground plane analysis, soldering
Power & Interfaces: HDMI, USB 2.0 hubs, USB-C data, DC to DC conversion, power distribution, supply supervision, reset circuitry
Programming & Tools: C, C++, Python, STM32F030, GNU Arm toolchain, GNU Make, Fusion 360, oscilloscopes, Git, Linux
Verification: Unit testing, model based testing, mutation testing, electrical/design rule checks, Gerber connectivity, BOM generation

EXPERIENCE

Guadaloop | Power Electronics Team Jan. 2026 to Present

- Designed Guadaloop power PCBs with buck converters and LDOs to step down a 60 V battery for onboard electronic subsystems requiring lower voltages and distribute 60 V directly to the levitation and fluids systems according to their power requirements.
- Assembled custom circuitry and an inverter to supply Guadaloop's propulsion system from the pod's 600 V battery source.
- Modeled a faraday cage enclosure to shield the pod's electronics and limit electromagnetic interference with nearby subsystems.
- Collaborated with members to work on the existing low voltage board to solder SMD components to mitigate with high frequencies
- Co-designed a four-layer, 60 V H-bridge PCB for magnetic levitation, integrating gate drivers, and bidirectional current sensing.

PROJECTS

HDMI/USB KVM Switch for Two Computers | KiCad, STM32F030, C Jun. 2026 to Aug. 2026

- Designed a 100 x 100 mm PCB layout with four layers in KiCad to switch two HDMI displays and a USB 2.0 hub with four ports between two computers, using differential pair routing and ground plane analysis to target 100 Ω HDMI and 90 Ω USB impedance.
- Integrated 3.3 V and 5 V supply supervisors with hardware reset gating and capacitor current limiting to hold the USB hub inactive during startup, microcontroller restarts, and supply faults, releasing it only when monitored voltages met operating requirements.
- Developed C firmware without an operating system for an STM32F030, implementing button debouncing, host detection, and watchdog recovery to coordinate HDMI and USB switching with a 150 ms disconnect interval followed by 100 ms of settling.
- Tested STM32 firmware across 804 modeled states and 173,664 transition cases and detected six injected defects through mutation testing, then checked PCB design rules and Gerber connectivity with zero design rule violations or unconnected board items.

Wireless Power Transfer | Independent research for Guadaloop - In progress Mar. 2026 to Present

- Reviewed multiple IEEE papers on wireless power transfer to understand how to implement the same system into the hyperloop.
- Selected series compensation on the transmitter and receiver coils for the initial 85 kHz design, using one capacitor per coil to tune resonance and simplify prototype construction compared with LCC networks that require additional inductors and capacitors.
- Built inverter and rectifier circuits for a moving wireless power transfer system, converting DC into AC to drive the transmitter coil and received AC back into DC for a pod supply intended to power onboard electronics, magnetic levitation, and braking subsystems.
- Investigated coil geometries, transmitter segmentation, and resonant compensation networks for a proposed 60 V supply and 12 foot track, comparing circuit complexity and sensitivity to coil misalignment to guide wireless power delivery to a moving pod.

LC-3b Processor Simulation & Virtual Memory | C Apr. 2026

- Built a C simulator for an LC-3b processor with five pipeline stages, modeling control and data latches and instruction cache latency over multiple cycles while implementing stalls without forwarding and instruction squashing for branches, jumps, and traps.
- Validated pipeline timing and final register and memory state against reference simulators using custom assembly tests for consecutive dependencies, dependencies after loads, and branch hazards, comparing stall and squash counts with expected cycle behavior.
- Extended a separate microcoded LC-3b simulator with virtual memory by adding address translation states and widening microinstructions for new control signals, handling exceptions like page fault, protection, unaligned, and illegal opcode exceptions.

KiCad Datasheet RAG Assistant | KiCad, Python, datasheet analysis Mar. 2026

- Linked KiCad netlists to datasheet retrieval for AI-assisted compatibility reviews, tracing pin and net connections and returning source pages for electrical limits while flagging missing documents and stale indexes after saved schematic or datasheet changes.
- Verified logic margin, pin voltage, regulator thermal, current budget, and I^2C pull-up checks against synthetic circuit failure cases.

Embedded Battleship System | C, ARM Cortex-M0+, PCB assembly Apr. 2025

- Programmed an MSPM0G3507 in C with a six-state machine and compact board data structures for a two-player Battleship game.
- Designed and soldered custom PCBs connecting the microcontroller to buttons, LCD, LEDs, and speaker, and programmed ST7735 graphics routines for bitmap rendering, grid displays, cursor control, and text in English and Spanish on a 160 x 110 pixel grid.